

The per-individual mixture-of-codings single-effect regression

Mean-field variational EM: model, free energy, and update equations

William R. P. Denault

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Abstract

We derive, from first principles, the mean-field variational EM algorithm implemented in `ser_pim_em` (package `MixSER`). The model is a single causal locus at which different individuals may act through different genetic coding models (additive, dominant, recessive); a latent per-individual assignment selects the coding, each coding carries one shared random effect, and a single intercept absorbs the trait baseline. We specify the variational family, write the evidence lower bound (ELBO / negative free energy) in closed form, and obtain every update — the responsibilities, the per-coding effect posterior, the mixing weights, the intercept, and the empirical-Bayes prior variances — as the exact block-coordinate maximiser of that single objective. Consequently the ELBO is non-decreasing at every step. We close with the two reasons the implementation uses an intercept rather than mean-centring, a note on identifiability at a single locus, and an appendix on the analytic single-coding (Level-1) locus Bayes factor that the routine also returns.

1 Model and notation

Index individuals by $i = 1, \dots, n$ and coding models by $m = 1, \dots, M$ (typically $M = 3$: additive, dominant, recessive). For coding m , let $x^{(m)} = (x_1^{(m)}, \dots, x_n^{(m)})^\top \in \mathbb{R}^n$ be the raw genotype encoding of the single locus under that model (e.g. additive 0/1/2, dominant $\mathbf{1}\{g \geq 1\}$, recessive $\mathbf{1}\{g = 2\}$). Let $y \in \mathbb{R}^n$ be the response (a partial residual when used inside SuSiE). The residual variance σ^2 is fixed externally (the outer IBSS loop updates it).

The generative model introduces a latent coding label $z_i \in \{1, \dots, M\}$ for each individual, one shared effect β_m per coding, and a shared intercept α :

$$z_i \sim \text{Cat}(\boldsymbol{\pi}), \quad \boldsymbol{\pi} = (\pi_1, \dots, \pi_M), \quad \sum_m \pi_m = 1, \quad (1)$$

$$\beta_m \sim \mathcal{N}(0, \sigma_{0m}^2), \quad m = 1, \dots, M, \quad (2)$$

$$y_i | z_i = m, \boldsymbol{\beta} \sim \mathcal{N}(\alpha + x_i^{(m)} \beta_m, \sigma^2). \quad (3)$$

Thus individual i “reads” the locus through its latent coding z_i , and all individuals sharing coding m share the *same* effect β_m . The unknown parameters are $\boldsymbol{\theta} = (\alpha, \boldsymbol{\pi}, \{\sigma_{0m}^2\}_m)$; the latent variables are $\mathbf{z} = (z_1, \dots, z_n)$ and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_M)$.

The joint density of observed and latent quantities factorises as

$$p(y, \mathbf{z}, \boldsymbol{\beta} | \boldsymbol{\theta}) = \underbrace{\prod_{m=1}^M \mathcal{N}(\beta_m; 0, \sigma_{0m}^2)}_{\text{priors on effects}} \prod_{i=1}^n \pi_{z_i} \mathcal{N}(y_i; \alpha + x_i^{(z_i)} \beta_{z_i}, \sigma^2). \quad (4)$$

The quantity of interest, the marginal likelihood $p(y | \boldsymbol{\theta}) = \sum_{\mathbf{z}} \int p(y, \mathbf{z}, \boldsymbol{\beta} | \boldsymbol{\theta}) d\boldsymbol{\beta}$, is intractable: it couples the M^n assignments with the M continuous effects. We therefore lower-bound it variationally.

2 Variational family and the free energy

Mean-field family. We approximate the posterior $p(\mathbf{z}, \boldsymbol{\beta} | y, \boldsymbol{\theta})$ by a fully factorised distribution

$$q(\mathbf{z}, \boldsymbol{\beta}) = \left(\prod_{i=1}^n q(z_i) \right) \left(\prod_{m=1}^M q(\beta_m) \right), \quad q(z_i) = \text{Cat}(\mathbf{r}_i), \quad q(\beta_m) = \mathcal{N}(\mu_m, s_m^2). \quad (5)$$

Here $r_{im} := q(z_i = m) \geq 0$ with $\sum_m r_{im} = 1$ are the *responsibilities*, and (μ_m, s_m^2) are the mean and variance of the Gaussian belief over the coding- m effect. We deliberately keep $q(\beta_m)$ Gaussian; §4 shows this family is closed under the update, so the restriction costs nothing.

ELBO. For any q the log marginal likelihood admits the decomposition $\log p(y | \boldsymbol{\theta}) = \mathcal{L}(q, \boldsymbol{\theta}) + \text{KL}(q \| p(\cdot | y, \boldsymbol{\theta}))$, where the evidence lower bound (ELBO; the negative variational free energy) is

$$\mathcal{L}(q, \boldsymbol{\theta}) = \mathbb{E}_q[\log p(y, \mathbf{z}, \boldsymbol{\beta} | \boldsymbol{\theta})] + \mathcal{H}(q), \quad \mathcal{H}(q) = -\mathbb{E}_q[\log q]. \quad (6)$$

Because $\text{KL} \geq 0$, $\mathcal{L} \leq \log p(y | \boldsymbol{\theta})$, and maximising $\mathcal{L}$ jointly over q and $\boldsymbol{\theta}$ is the variational EM problem.

Closed form. Substituting (4) and (5) into (6) and using $\sum_m r_{im} = 1$, the entropy of the factorised q , and $\mathbb{E}_{q(\beta_m)}[\beta_m] = \mu_m$, $\mathbb{E}_{q(\beta_m)}[\beta_m^2] = \mu_m^2 + s_m^2$, gives

$$\mathcal{L} = \sum_{m=1}^M \sum_{i=1}^n r_{im} \left(\log \pi_m + \ell_{im} - \log r_{im} \right) - \sum_{m=1}^M \text{KL}(\mathcal{N}(\mu_m, s_m^2) \| \mathcal{N}(0, \sigma_{0m}^2)), \quad (7)$$

where the per-pair expected log-likelihood is

$$\ell_{im} := \mathbb{E}_{q(\beta_m)}[\log \mathcal{N}(y_i; \alpha + x_i^{(m)} \beta_m, \sigma^2)] = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \left[(y_i - \alpha - x_i^{(m)} \mu_m)^2 + (x_i^{(m)})^2 s_m^2 \right], \quad (8)$$

and the Gaussian–Gaussian Kullback–Leibler divergence is the standard

$$\text{KL}(\mathcal{N}(\mu_m, s_m^2) \| \mathcal{N}(0, \sigma_{0m}^2)) = \frac{1}{2} \left[\log \frac{\sigma_{0m}^2}{s_m^2} + \frac{s_m^2 + \mu_m^2}{\sigma_{0m}^2} - 1 \right]. \quad (9)$$

Equations (7)–(9) are exactly the ELBO accumulated in the implementation. The derivation of (8) uses only $\mathbb{E}[(y_i - \alpha - x_i^{(m)} \beta_m)^2] = (y_i - \alpha - x_i^{(m)} \mu_m)^2 + (x_i^{(m)})^2 \text{Var}_q(\beta_m)$ with $\text{Var}_q(\beta_m) = s_m^2$.

The algorithm maximises the *single* objective (7) by cyclic block-coordinate ascent over the blocks $\{q(z_i)\}$, $\{q(\beta_m)\}$, $\boldsymbol{\pi}$, α , $\{\sigma_{0m}^2\}$. We derive each block’s maximiser in turn; since every block is solved exactly, $\mathcal{L}$ never decreases (§6).

3 E-step: responsibilities r_{im}

Fix θ and $\{q(\beta_m)\}$ and maximise $\mathcal{L}$ over a single $q(z_i)$ subject to $\sum_m r_{im} = 1$. Only the first sum in (7) depends on $q(z_i)$:

$$\mathcal{L}(q(z_i)) = \sum_{m=1}^M r_{im} (\log \pi_m + \ell_{im} - \log r_{im}) + \text{const.} \quad (10)$$

Introducing a Lagrange multiplier λ for the constraint and setting $\partial/\partial r_{im} = 0$,

$$\log \pi_m + \ell_{im} - \log r_{im} - 1 - \lambda = 0 \implies r_{im} \propto \pi_m e^{\ell_{im}}. \quad (11)$$

This is the generic mean-field fixed point $q^*(z_i) \propto \exp \mathbb{E}_{q(\beta)}[\log p(y, z, \beta)]$. Dropping the constant $-\frac{1}{2} \log(2\pi\sigma^2)$ from (8) (it cancels in the normalisation) yields the implemented form

$$r_{im} = \frac{\pi_m \exp(\eta_{im})}{\sum_{k=1}^M \pi_k \exp(\eta_{ik})}, \quad \eta_{im} = -\frac{1}{2\sigma^2} \left[(y_i - \alpha - x_i^{(m)} \mu_m)^2 + (x_i^{(m)})^2 s_m^2 \right]. \quad (12)$$

The effect is integrated against its *current* belief $q(\beta_m)$: the score η_{im} compares individual i 's observation to the fitted value $\alpha + x_i^{(m)} \mu_m$ (never to 0), penalised by the predictive uncertainty $(x_i^{(m)})^2 s_m^2$. Individuals are thus assigned to codings by fit quality, with a built-in Occam penalty for uncertain effects.

Remark 1 (Reference genotype is uninformative). For an individual with $g_i = 0$ every coding gives $x_i^{(m)} = 0$, so $\eta_{im} = -(y_i - \alpha)^2 / (2\sigma^2)$ is the *same* for all m and $r_{im} = \pi_m$. Such individuals carry no coding information and correctly revert to the prior. This is exactly why the baseline must enter as the intercept α and not by mean-centring y (§7).

4 M-step I: per-coding effect posterior $q(\beta_m)$

Fix $\{r_{im}\}$ and θ and optimise over $q(\beta_m)$. The generic mean-field optimum is $\log q^*(\beta_m) = \mathbb{E}_{q(z)}[\log p(y, z, \beta)] + \text{const}$, where the expectation is over everything except β_m . Collecting the β_m -dependent terms of (4),

$$\log q^*(\beta_m) = -\frac{1}{2\sigma^2} \sum_{i=1}^n r_{im} (y_i - \alpha - x_i^{(m)} \beta_m)^2 - \frac{\beta_m^2}{2\sigma_{0m}^2} + \text{const.} \quad (13)$$

This is quadratic in β_m , so $q^*(\beta_m)$ is Gaussian (the family is closed). Matching the coefficients of β_m^2 and β_m gives the precision and mean:

$$\frac{1}{s_m^2} = \frac{1}{\sigma_{0m}^2} + \frac{1}{\sigma^2} \sum_{i=1}^n r_{im} (x_i^{(m)})^2, \quad (14)$$

$$\mu_m = \frac{s_m^2}{\sigma^2} \sum_{i=1}^n r_{im} x_i^{(m)} (y_i - \alpha). \quad (15)$$

Writing $S_m := \sum_i r_{im} (x_i^{(m)})^2$, these are the boxed updates

$$s_m^2 = \left(\frac{1}{\sigma_{0m}^2} + \frac{S_m}{\sigma^2} \right)^{-1}, \quad \mu_m = \frac{s_m^2}{\sigma^2} \sum_{i=1}^n r_{im} x_i^{(m)} (y_i - \alpha). \quad (16)$$

This is precisely a responsibility-weighted Bayesian single-effect regression of $y - \alpha$ on coding m : each individual contributes with weight r_{im} , and the prior $\mathcal{N}(0, \sigma_{0m}^2)$ shrinks the estimate.

5 M-step II: hyperparameters

5.1 Mixing weights π

Only $\sum_m (\sum_i r_{im}) \log \pi_m$ in (7) depends on $\boldsymbol{\pi}$. Maximising under $\sum_m \pi_m = 1$ with a Lagrange multiplier gives the ordinary EM weight update

$$\pi_m = \frac{1}{n} \sum_{i=1}^n r_{im}. \quad (17)$$

No `mixsqp` / empirical-Bayes step is applied to the per-individual responsibilities; doing so collapses the weights onto the single sparsest coding (see §7).

5.2 Intercept α

Differentiating (7) through (8),

$$\frac{\partial \mathcal{L}}{\partial \alpha} = \frac{1}{\sigma^2} \sum_{i=1}^n \sum_{m=1}^M r_{im} (y_i - \alpha - x_i^{(m)} \mu_m) = 0. \quad (18)$$

Using $\sum_m r_{im} = 1$ and solving,

$$\alpha = \frac{1}{n} \sum_{i=1}^n \left(y_i - \sum_{m=1}^M r_{im} x_i^{(m)} \mu_m \right). \quad (19)$$

The intercept is the mean residual after removing the responsibility-averaged mixed effect.

5.3 Empirical-Bayes prior variances σ_{0m}^2

The only σ_{0m}^2 -dependent part of (7) is $-\text{KL}(\mathcal{N}(\mu_m, s_m^2) \parallel \mathcal{N}(0, \sigma_{0m}^2))$. Holding $q(\beta_m)$ fixed and writing $v := \sigma_{0m}^2$,

$$\frac{\partial}{\partial v} \left[-\frac{1}{2} \log v - \frac{s_m^2 + \mu_m^2}{2v} \right] = -\frac{1}{2v} + \frac{s_m^2 + \mu_m^2}{2v^2} = 0 \implies v = s_m^2 + \mu_m^2. \quad (20)$$

Hence the ELBO-maximising empirical-Bayes update is the second moment of the effect belief,

$$\sigma_{0m}^2 = \mu_m^2 + s_m^2 = \mathbb{E}_{q(\beta_m)}[\beta_m^2] \quad (21)$$

(applied when `estimate_prior_variance=TRUE`; a floor 10^{-8} keeps it positive). Equations (14) and (21) are coupled across iterations but each is the exact maximiser of its own block, which is all coordinate ascent requires.

6 Monotonicity, the algorithm, and convergence

Proposition 1 (Monotone ELBO). *Each update (12), (14)–(15), (17), (19), (21) is the exact global maximiser of the single objective $\mathcal{L}$ in (7) over its block, holding the others fixed. Therefore the sequence of ELBO values produced by cyclic application is non-decreasing, $\mathcal{L}^{(t+1)} \geq \mathcal{L}^{(t)}$, and, being bounded above by $\log p(y \mid \boldsymbol{\theta})$, it converges.*

Proof. Block-coordinate ascent on a function with exact block maximisers is monotone: replacing one block by its argmax cannot decrease the objective, and the other blocks are unchanged. The $q(z_i)$ and $q(\beta_m)$ steps are the mean-field optima (§3,§4); the π , α , σ_{0m}^2 steps are the stationary points of concave one-block subproblems (§5). The bound $\mathcal{L} \leq \log p(y|\theta)$ follows from $\text{KL} \geq 0$. A monotone sequence bounded above converges. $\square$

Algorithm (one sweep). Given σ^2 , repeat to convergence of $\mathcal{L}$:

1. **E-step:** update all r_{im} by (12);
2. **weights:** $\pi_m \leftarrow \frac{1}{n} \sum_i r_{im}$ (17);
3. **intercept:** α by (19);
4. **effect:** s_m^2, μ_m by (14)–(15);
5. **EB prior:** $\sigma_{0m}^2 \leftarrow \mu_m^2 + s_m^2$ (21).

Initialise π uniform, $\alpha = \bar{y}$, and each μ_m at the single-coding OLS slope of $y - \alpha$ on $x^{(m)}$ with $s_m^2 = \sigma^2 / \|x^{(m)}\|^2$. Because the per-individual mixture is only weakly identified at one locus (§7), the routine runs `n_start` restarts — the first uniform, the rest Dirichlet(1) draws of π — and returns the fit of highest ELBO.

Variational marginal likelihood. A convenient by-product evaluated during the E-step is

$$\tilde{\ell}(y) = \sum_{i=1}^n \log \sum_{m=1}^M \pi_m e^{\ell_{im}}, \quad (22)$$

the log of the (effect-integrated) per-individual mixture likelihood at the current beliefs; it is reported as `loglik`.

7 Why an intercept, not mean-centring; identifiability

Intercept vs. centring. By Remark 1, raw 0/1/2 codings make the reference genotype ($g_i = 0$) predict α under every coding, so such individuals are uninformative ($r_{im} = \pi_m$). If instead one mean-centres y and drops α , the $g = 0$ group sits at $y \approx -\bar{y}$ while the model predicts 0 there; every component must then bend to absorb the baseline, and the optimiser collapses onto whichever coding can do so most cheaply — the spurious “recessive collapse”. The shared intercept (19) removes the baseline without this distortion and is invariant to the trait mean.

Weights by EM, not mixsqp. Treating the columns $\{r_{.m}\}$ as data for an empirical-Bayes mixing solver (`mixsqp`) maximises a different objective that rewards sparsity in π and again collapses to one coding. The plain EM update (17) is the correct M-step for (7).

Identifiability at one locus. The dominant coding $\mathbf{1}\{g \geq 1\}$ equals the additive coding except at $g = 2$, and the recessive coding $\mathbf{1}\{g = 2\}$ is informative only at $g = 2$; the contrasts that separate codings live entirely in the $g \geq 1$ stratum, and $g = 0$ individuals are uninformative. A single locus therefore identifies the per-individual mixture only weakly, and a residual dominant component can persist. The restarts guard the optimisation, but *calibrated* weights π require pooling the responsibilities across many loci; at one locus the quantity to trust for a decision is the analytic Level-1 Bayes factor of Appendix A, not π itself.

A Analytic single-coding (Level-1) locus Bayes factor

Independently of the per-individual mixture, `ser_pim_em` also returns the closed-form Bayes factor of “some coding has a non-zero effect” versus “all effects are zero”. For each coding m let $\tilde{x}^{(m)}$ be the centred-and-scaled $x^{(m)}$ and $\tilde{y} = y - \bar{y}$. The exact single-effect Bayes factor of Wang et al. (2020) for coding m at prior variance V is

$$\log \text{BF}_m(V) = \frac{1}{2} \log \frac{\hat{s}_m^2}{V + \hat{s}_m^2} + \frac{\hat{\beta}_m^2}{2} \left(\frac{1}{\hat{s}_m^2} - \frac{1}{V + \hat{s}_m^2} \right), \quad \hat{\beta}_m = \frac{\tilde{x}^{(m)\top} \tilde{y}}{\|\tilde{x}^{(m)}\|^2}, \quad \hat{s}_m^2 = \frac{\sigma^2}{\|\tilde{x}^{(m)}\|^2}. \quad (23)$$

Each V_m is set by maximum marginal likelihood, $V_m = \arg \max_{V \geq 0} \log \text{BF}_m(V)$ (with the off-switch $V_m = 0$, i.e. $\log \text{BF}_m = 0$, whenever the optimum is negative). With coding weights w_m (uniform $1/M$ by default) the locus Bayes factor of any-coding-effect vs. null is the weighted average

$$\log \text{BF}_{\text{locus}} = \log \sum_{m=1}^M w_m \text{BF}_m(V_m) = \log \sum_m \exp(\log w_m + \log \text{BF}_m(V_m)), \quad (24)$$

returned as `lbf_snp/bf_snp`. This is the object the SuSiE / SuSiE 2.0 IBSS loop consumes as the single-effect log-Bayes-factor for this locus; the per-individual mixture’s exact Bayes factor would require the intractable M^n sum, so for decisions the analytic (24) is used. As a conservative cross-check the routine also reports the variational lower bound $\log \text{BF}_{\text{pim}} = \mathcal{L} - \ell_0$, where $\ell_0 = \sum_i \log \mathcal{N}(y_i; \bar{y}, \sigma^2)$ is the intercept-only null log-likelihood.

Reference. G. Wang, A. Sarkar, P. Carbonetto & M. Stephens (2020). *A simple new approach to variable selection in regression, with application to genetic fine mapping*. JRSS-B 82(5), 1273–1300.